

PAPER_683: UQFF Hawking Temperature Modulation: Spectral Shift and Wien Peak Correction

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Framework: Unified Quantum Field Framework (UQFF) v5.30

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Class: UQFFHawkingTemperatureModulation | CP4 #267

Source: grok_share_fc21e30c24b4.txt

Domain: Hawking Radiation / Black Hole Thermodynamics

Abstract

The standard Hawking temperature $T_H = \hbar c^3 / (8\pi G M k_B)$ is modulated in UQFF through three channels: time-reversal boost $(1 + f_{\text{TRZ}})$, condensate suppression $(1 - \rho_{\text{SCm}}/\rho_{\text{UA}})$, and magnetic string correction $(1 + U_m/k_B T_H)$. The combined modulation shifts the Planck spectrum peak by $\Delta\lambda_{\text{max}} = \hbar c / (2.82 k_B) (1/T_{\text{UQFF}} - 1/T_H)$, observable in principle for primordial micro-BHs.

Primary UQFF Equation

$$T_{\text{UQFF}} = T_H (1 + f_{\text{TRZ}}) \left(1 - \frac{\rho_{\text{SCm}}}{\rho_{\text{UA}}}\right) \left(1 + \frac{U_m}{k_B T_H}\right)$$

Parameters

Factor	Value	Effect
$1 + f_{\text{TRZ}}$	1.1	Boost
$1 - \rho_{\text{SCm}}/\rho_{\text{UA}}$	0.9	Suppression
$1 + U_m/k_B T_H$	~ 1	Modulation

UQFF Constants

Constant	Value	Description
ρ_{UA}	$7.09 \times 10^{-36} \text{ kg/m}^3$	Universal Aether density
ρ_{SCm}	$7.09 \times 10^{-37} \text{ kg/m}^3$	Schwarzschild condensate
f_{TRZ}	0.1	Time-reversal zone factor

Constant	Value	Description
κ	$5 \times 10^{-4} \text{ day}^{-1}$	UQFF calibration constant
[SSq]	0.57	Superstring quenching factor
μ_J	$3.38 \times 10^{23} \text{ J}\cdot\text{m}$	Magnetic string coupling
γ	$5 \times 10^{-5} / 86400 \text{ s}^{-1}$	Aether oscillation decay

C++ Implementation

```
#include "UQFFHawkingTemperatureModulation.h"
```

```
UQFFHawkingTemperatureModulation obj;
auto result = obj.compute_primary(...); // primary equation
obj.simulate_over_M(M_start, M_end, dM, "output.csv");
```

Python CP4 Calculator

```
from CondensedPhysics2 import UQFFHawkingTemperatureModulationCalculator

calc = UQFFHawkingTemperatureModulationCalculator()
result = calc.compute() # returns all UQFF quantities
print(result)
```

References

Hawking 1974; Bekenstein 1973; UQFF Session 173

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